
Chapter 8

Confidence Intervals and Sample Size

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Outline

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Objectives

- Find the confidence interval for the mean when σ is known or $n \geq 30$.
- Determine the minimum sample size for finding a confidence interval for the mean.

Objectives

- Find the confidence interval for the mean when σ is unknown and $n < 30$.
- Find the confidence interval for a proportion.
- Determine the minimum sample size for finding a confidence interval for a proportion.
- Find a confidence interval for a variance and a standard deviation.

8-2 Confidence Intervals for the Mean (σ Known or $n \geq 30$) and Sample Size

A point estimate is a specific numerical value estimate of a parameter. The best estimate of the population mean μ is the sample mean $\bar{X}$.

8-2 Three Properties of a Good Estimator

- The estimator must be an **unbiased estimator**. That is, the expected value or the mean of the estimates obtained from samples of a given size is equal to the parameter being estimated.

8-2 Three Properties of a Good Estimator

- The estimator must be consistent. For a **consistent estimator**, as sample size increases, the value of the estimator approaches the value of the parameter estimated.

8-2 Three Properties of a Good Estimator

- The estimator must be a **relatively efficient estimator**. That is, of all the statistics that can be used to estimate a parameter, the relatively efficient estimator has the smallest variance.

8-2 Confidence Intervals

- An **interval estimate** of a parameter is an interval or a range of values used to estimate the parameter. This estimate may or may not contain the value of the parameter being estimated.

8-2 Confidence Intervals

- A **confidence interval** is a specific interval estimate of a parameter determined by using data obtained from a sample and the specific confidence level of the estimate.

8-2 Confidence Intervals

- The **confidence level** of an interval estimate of a parameter is the probability that the interval estimate will contain the parameter.

8-2 Formula for the Confidence Interval of the Mean for a Specific α

- The **confidence level** is the percentage equivalent to the decimal value of $1 - \alpha$.

$$\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

8-2 Maximum Error of Estimate

- The **maximum error of estimate** is the maximum difference between the point estimate of a parameter and the actual value of the parameter.

8-2 Confidence Intervals - Example

- The president of a large university wishes to estimate the average age of the students presently enrolled. From past studies, the standard deviation is known to be 2 years. A sample of 50 students is selected, and the mean is found to be 23.2 years. Find the 95% confidence interval of the population mean.

8-2 Confidence Intervals - Example

Since the 95% confidence interval is desired, $z_{\alpha/2} = 1.96$. Hence, substituting in the formula

$$\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

one gets

8-2 Confidence Intervals - Example

$$23.2 - (1.96)\left(\frac{2}{\sqrt{50}}\right) < \mu < 23.2 + (1.96)\left(\frac{2}{\sqrt{50}}\right)$$

$$23.2 - 0.6 < \mu < 23.2 + 0.6$$

$$22.6 < \mu < 23.8 \text{ or } 23.2 \pm 0.6 \text{ years.}$$

Hence, the president can say, with 95% confidence, that the average age of the students is between 22.6 and 23.8 years, based on 50 students.

8-2 Confidence Intervals - Example

- A certain medication is known to increase the pulse rate of its users. The standard deviation of the pulse rate is known to be 5 beats per minute. A sample of 30 users had an average pulse rate of 104 beats per minute. Find the 99% confidence interval of the true mean.

8-2 Confidence Intervals - Example

Since the 99% confidence interval is desired, $z_{\alpha/2} = 2.58$. Hence, substituting in the formula

$$\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

one gets

8-2 Confidence Intervals - Example

$$104 - (2.58)\left(\frac{5}{\sqrt{30}}\right) < \mu < 104 + (2.58)\left(\frac{5}{\sqrt{30}}\right)$$

$$104 - 2.4 < \mu < 104 + 2.4$$

$$101.6 < \mu < 106.4.$$

Hence, one can say, with 99% confidence, that the average pulse rate is between 101.6 and 106.4 beats per minute, based on 30 users.

8-2 Formula for the Minimum Sample Size Needed for an Interval Estimate of the Population Mean

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

where E is the maximum error of estimate.

If necessary, round the answer up to obtain a whole number.

8-2 Minimum Sample Size Needed for an Interval Estimate of the Population Mean - **Example**

- The college president asks the statistics teacher to estimate the average age of the students at their college. How large a sample is necessary? The statistics teacher decides the estimate should be accurate within 1 year and be 99% confident. From a previous study, the standard deviation of the ages is known to be 3 years.

8-2 Minimum Sample Size Needed for an Interval Estimate of the Population Mean - **Example**

*Since $\alpha = 0.01$ (or $1 - 0.99$),
 $z_{\alpha/2} = 2.58$, and $E = 1$, substituting*

in $n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E} \right)^2$ gives

$$n = \left(\frac{(2.58)(3)}{1} \right)^2 = 59.9 \approx 60.$$

8-3 Characteristics of the *t* Distribution

- The *t* distribution shares some characteristics of the normal distribution and differs from it in others. The *t* distribution is similar to the standard normal distribution in the following ways:
 - It is bell-shaped.
 - It is symmetrical about the mean.

8-3 Characteristics of the *t* Distribution

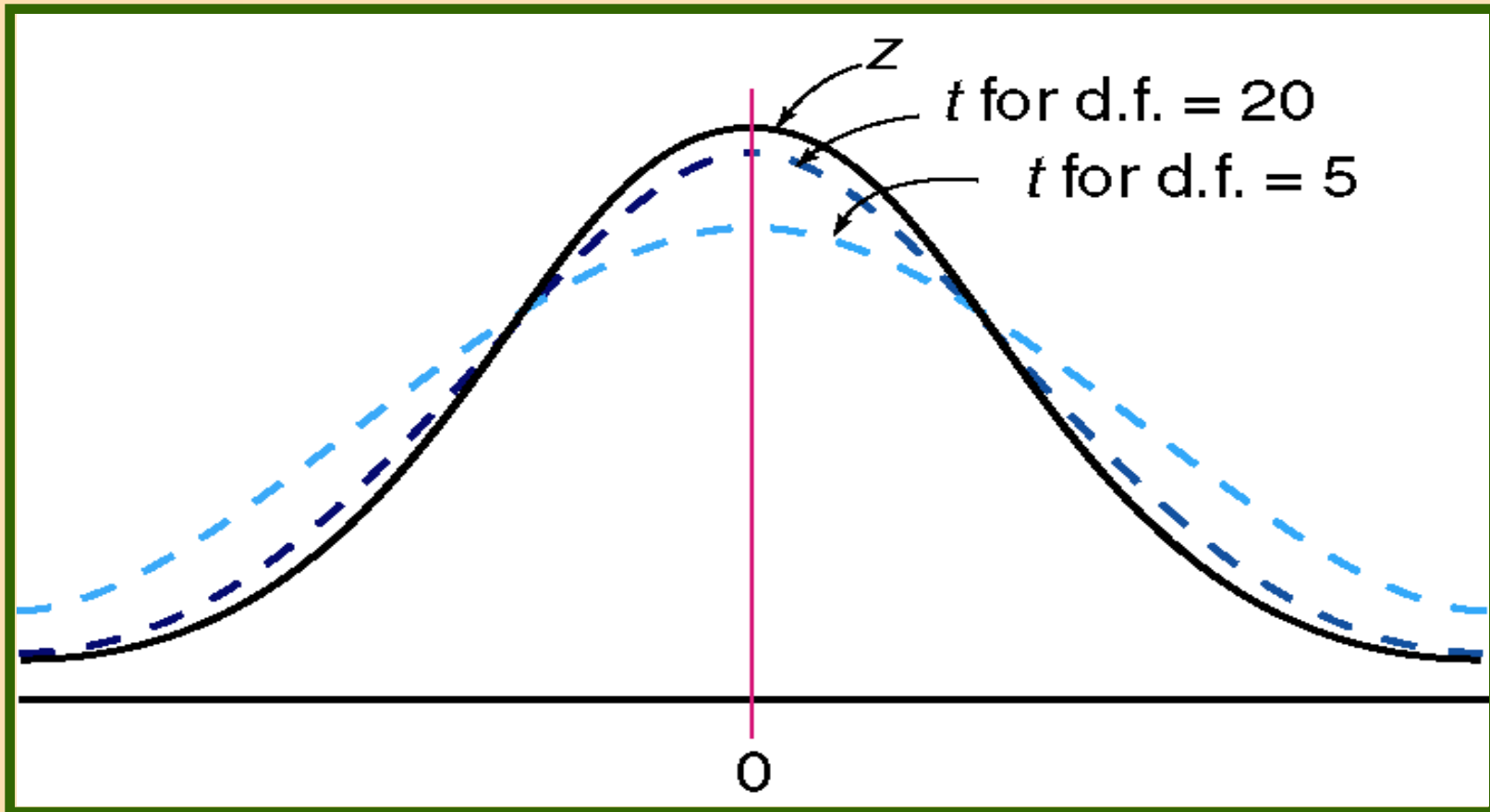
- The mean, median, and mode are equal to 0 and are located at the center of the distribution.
- The curve never touches the x axis.
- The *t* distribution differs from the standard normal distribution in the following ways:

8-3 Characteristics of the *t* Distribution

- The variance is greater than 1.
- The *t* distribution is actually a family of curves based on the concept of *degrees of freedom*, which is related to the sample size.
- As the sample size increases, the *t* distribution approaches the standard normal distribution.

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8-3 Standard Normal Curve and the t Distribution



8-3 Confidence Interval for the Mean (σ Unknown and $n < 30$) - Example

- Ten randomly selected automobiles were stopped, and the tread depth of the right front tire was measured. The mean was 0.32 inch, and the standard deviation was 0.08 inch. Find the 95% confidence interval of the mean depth. Assume that the variable is approximately normally distributed.

8-3 Confidence Interval for the Mean (σ Unknown and $n < 30$) - Example

- Since σ is unknown and s must replace it, the t distribution must be used with $\alpha = 0.05$. Hence, with 9 degrees of freedom, $t_{\alpha/2} = 2.262$ (see Table F in text).
- From the next slide, we can be 95% confident that the population mean is between 0.26 and 0.38.

8-3 Confidence Interval for the Mean (σ Unknown and $n < 30$) - Example

*Thus the 95% confidence interval
of the population mean is found by
substituting in*

$$\bar{X} - t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{X} + t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

$$0.32 - (2.262) \left(\frac{0.08}{\sqrt{10}} \right) < \mu < 0.32 + (2.262) \left(\frac{0.08}{\sqrt{10}} \right)$$

$$0.26 < \mu < 0.38$$

8-4 Confidence Intervals and Sample Size for Proportions

Symbols Used in Proportion Notation

p = population proportion

$\hat{p}$ (read “ p – hat”) = sample proportion

$$\hat{p} = \frac{X}{n} \text{ and } \hat{q} = \frac{n - X}{n} \text{ or } 1 - \hat{p}$$

where X = number of sample units that possess the characteristic of interest and n = sample size.

8-4 Confidence Intervals and Sample Size for Proportions - **Example**

- In a recent survey of 150 households, 54 had central air conditioning. Find $\hat{p}$ and $\hat{q}$.

8-4 Confidence Intervals and Sample Size for Proportions - **Example**

Since $X = 54$ and $n = 150$, then

$$\hat{p} = \frac{X}{n} = \frac{54}{150} = 0.36 = 36\%$$

$$\begin{aligned} \text{and } \hat{q} &= \frac{n - X}{n} = \frac{150 - 54}{150} = \frac{96}{150} \\ &= 0.64 = 64\% \end{aligned}$$

$$\text{or } \hat{q} = 1 - \hat{p} = 1 - 0.36 = 0.64.$$

8-4 Formula for a Specific Confidence Interval for a Proportion

$$\hat{p} - (z_{\alpha/2})\sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + (z_{\alpha/2})\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

8-4 Specific Confidence Interval for a Proportion - Example

- A sample of 500 nursing applications included 60 from men. Find the 90% confidence interval of the true proportion of men who applied to the nursing program.
- Here $\alpha = 1 - 0.90 = 0.10$, and $z_{\alpha/2} = 1.65$.
- $\hat{p} = 60/500 = 0.12$ and $\hat{q} = 1 - 0.12 = 0.88$.

8-4 Specific Confidence Interval for a Proportion - Example

Substituting in

$$\hat{p} - (z_{\alpha/2}) \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + (z_{\alpha/2}) \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

we get

$$\text{Lower limit} = 0.12 - (1.65) \sqrt{\frac{(0.12)(0.88)}{500}} = 0.096$$

$$\text{Upper limit} = 0.12 + (1.65) \sqrt{\frac{(0.12)(0.88)}{500}} = 0.144$$

Thus, $0.096 < p < 0.144$ or $9.6\% < p < 14.4\%$.

8-4 Sample Size Needed for Interval Estimate of a Population Proportion

$$n = \hat{p}\hat{q}\left(\frac{z_{\alpha/2}}{E}\right)^2$$

where E is the maximum error of estimate.

If necessary, round the answer up to obtain a whole number.

8-4 Sample Size Needed for Interval Estimate of a Population Proportion - **Example**

- **A researcher wishes to estimate, with 95% confidence, the number of people who own a home computer. A previous study shows that 40% of those interviewed had a computer at home. The researcher wishes to be accurate within 2% of the true proportion. Find the minimum sample size necessary.**

8-4 Sample Size Needed for Interval Estimate of a Population Proportion - **Example**

Since $\alpha = 0.05$, $z_{\alpha/2} = 1.96$, $E = 0.02$ $\hat{p} = 0.40$,

and $\hat{q} = 0.60$, then $n = \hat{p}\hat{q}\left(\frac{z_{\alpha/2}}{E}\right)^2$

$$= (0.40)(0.60)\left(\frac{1.96}{0.02}\right)^2 = 2304.96$$

Which, when rounded up is 2305 people to interview.

8-5 Confidence Intervals for Variances and Standard Deviations

- To calculate these confidence intervals, the **chi-square** distribution is used.
- The chi-square distribution is similar to the t distribution in that its distribution is a family of curves based on the number of degrees of freedom.
- The symbol for chi-square is χ^2 .

8-5 Confidence Interval for a Variance

Formula for the confidence interval for a variance

$$\frac{(n-1)s^2}{\chi^2_{\text{right}}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\text{left}}}$$

$$\text{d. f.} = n - 1$$

8-5 Confidence Interval for a Standard Deviation

Formula for the confidence interval for a standard deviation

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\text{right}}}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi^2_{\text{left}}}}$$

$$\text{d.f.} = n - 1$$

8-5 Confidence Interval for the Variance - Example

- Find the 95% confidence interval for the variance and standard deviation of the nicotine content of cigarettes manufactured if a sample of 20 cigarettes has a standard deviation of 1.6 milligrams.
- Since $\alpha = 0.05$, the critical values for the 0.025 and 0.975 levels for 19 degrees of freedom are 32.852 and 8.907.

8-5 Confidence Interval for the Variance - Example

The 95% confidence interval for the variance is found by substituting in

$$\frac{(n-1)s^2}{\chi^2_{\text{right}}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\text{left}}}$$

$$\frac{(20-1)(1.6)^2}{32.852} < \sigma^2 < \frac{(20-1)(1.6)^2}{8.907}$$

$$1.5 < \sigma^2 < 5.5$$

8-5 Confidence Interval for the Standard Deviation - Example

The 95% confidence interval for the standard deviation is

$$\sqrt{1.5} < \sigma < \sqrt{5.5}$$

$$1.2 < \sigma < 2.3$$